Dataset & Problem Proposal: Saaf Hawa v2

Air Quality Track | Zero-Cost Infrastructure

1. Problem Statement & Operational User

Following the March 2025 shutdown of U.S. Diplomatic Post AirNow monitors in Pakistan, public health and education authorities lost all continuous reference particulate data. District Education Authorities (DEAs) in Punjab must make critical operational decisions without trustworthy next-day forecasts. Saaf Hawa v2 reconstructs ground truth by calibrating community sensor networks against the pre-halt reference record and produces 24h uncertainty-bounded quantile forecasts.

2. Original Publishers & Multi-Source Independence

1. **Live Target Ground Truth:** Pakistan Air Quality Initiative (PAQI) via OpenAQ (CC BY 4.0 / Open Data).
2. **Historical Calibration Reference:** U.S. Diplomatic Post AirNow monitors via OpenAQ v3 (Public Domain). *[Source not on track list]*.
3. **Model Atmospheric Feed:** Copernicus ECMWF CAMS and ERA5 via Open-Meteo (CC BY 4.0 Non-commercial).
4. **Transboundary Fire Signal:** NASA Earthdata / LANCE FIRMS VIIRS & MODIS (Public Domain).
5. **Standards & Guidance:** WHO Global Air Quality Guidelines 2021 & Punjab Environmental Quality Standards (PEQS).

3. Storage Contract, Provenance & Baselines

• **Contract:** Timestamps stored in UTC. Missing target values remain NULL. Deterministic row IDs: {CITY}-{YYYYMMDDTHHZ}.
• **Volume:** 56,388 ground-truth city-hourly records, 28,652 embassy calibration rows, and 133,440 atmospheric feature rows, exceeding the 50,000 threshold.
• **Baselines:** Baseline A (Persistence at t) and Baseline B (Seasonal-Hour mean of target hour over past 7 days).
• **Primary Limitation:** Post-March 2025 ground truth is an operational estimate calibrated from community sensors rather than direct BAM reference measurements.
• **Integrity:** SHA-256 computed on ingest and audited in manifest.csv. 24-hour embargo enforced between splits.